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Enhancing Seismic Feature Detection Through 3D Gradient-Fused Shaded Relief and Dynamic Parameter Optimization

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Abstract

This paper aims to enhance feature detection in geophysical seismic data by advancing Arthur E. Barnes's 2002 shaded relief method. Our primary objective is to integrate a 3D gradient operator into the existing workflow, thereby improving the clarity of geological structures and enhancing interpretability across the entire seismic volume.

Building upon Barnes's method, which utilizes dip-azimuth attributes and directional illumination, our approach introduces several methodological improvements. We compute amplitude-derived gradients at the voxel level using a 3D gradient operator and incorporate these gradient vectors into the shaded relief algorithm. This integration enhances the visibility of subtle structural and stratigraphic features by improving illumination cues. Furthermore, we dynamically optimize "degree" and "azimuth" parameters, allowing for more adaptive illumination based on the characteristics of each slice. Finally, we merge the original seismic amplitude information with the shaded relief output, resulting in a more comprehensive and geologically interpretable volumetric representation.

Our enhanced technique demonstrates superior effectiveness in highlighting both structural and stratigraphic features compared to Barnes's original method. By integrating the 3D gradient operator, we reveal subtle variations in fault planes and reflectors, leading to improved clarity and interpretability. Applications on field datasets illustrate how dynamic parameter optimization—combined with gradient-based shading and subsequent data fusion—sharpens fault delineation and emphasizes complex geological details that were previously indistinct. This outcome underscores the value of tailoring illumination parameters to localized features, leading to a deeper understanding of subsurface morphology. Overall, the proposed framework provides a streamlined and robust means for advanced seismic volume interpretation.

The equipment used in this research includes a high-performance computer with advanced graphics capabilities, seismic data processing software, and a 3D visualization system. These tools were essential for implementing the 3D gradient operator and dynamic parameter optimization. The high-performance computer handled the complex computations involved in processing seismic data, while the seismic data processing software facilitated the integration of the enhanced algorithm into the workflow. The 3D visualization system allowed for detailed analysis and interpretation of the seismic volume. Testing involved applying the method to various seismic datasets to evaluate its effectiveness in different geological

settings. An unusual aspect of our approach was the dynamic optimization of illumination parameters, which required iterative testing and refinement to achieve optimal results. The equipment proved highly effective in handling the complex computations involved, demonstrating its suitability for advanced seismic interpretation tasks.

This study contributes a novel gradient-based enhancement to Barnes's shaded relief method, with dynamically optimized illumination parameters and a data-fusion approach. These innovations offer superior delineation of subsurface features, boosting the existing body of knowledge for fault detection and geological interpretation in the petroleum industry. The integration of advanced equipment and methodologies underscores the potential for continued advancements in seismic data analysis, ultimately contributing to more accurate and efficient energy exploration.

Keywords: Shaded relief, Gradient Amplitude, Fusion, Gradient-enhanced Visualization

Introduction

The existing method for feature detection in geophysical seismic data, known as the shaded relief method, has limitations in terms of clarity and interpretability of geological structures. This method, developed by Arthur E. Barnes in 2002, uses dip-azimuth attributes and directional illumination to enhance the visibility of geological features. However, it struggles to reveal subtle variations in fault planes and reflectors, making it difficult to accurately interpret subsurface morphology.

The lack of clarity and interpretability can lead to incorrect or incomplete understanding of geological features, which can have significant implications for the petroleum industry. For instance, inaccurate identification of fault planes and reflectors can lead to incorrect placement of wells, resulting in reduced hydrocarbon production and increased exploration costs. Furthermore, the inability to accurately interpret subsurface morphology can also lead to incorrect assessment of reservoir quality, which can impact the overall viability of a hydrocarbon project.

To address the limitations of the existing shaded relief method, a novel approach has been developed that integrates a 3D gradient operator into the original workflow. This approach enhances the illumination of subtle structural and stratigraphic features, resulting in improved clarity and interpretability of geological structures.

The 3D gradient operator computes amplitude-derived gradients at the voxel level, which are then injected into the shaded relief algorithm to accentuate subtle features. This integration allows for more effective highlighting of fault planes, reflectors, and other geological features, resulting in a more comprehensive and geologically interpretable volumetric representation.

In addition to the 3D gradient operator, the proposed solution also dynamically optimizes "degree" and "azimuth" parameters to allow for more adaptive illumination based on each slice's characteristics. This dynamic optimization enables the method to adapt to varying geological conditions, resulting in more accurate and detailed representation of subsurface morphology.

A data-fusion step is also implemented, which merges the original seismic amplitude information with the shaded relief output. This fusion of data results in a more comprehensive and geologically interpretable volumetric representation, allowing for more accurate identification of geological features and improved understanding of subsurface morphology.

The proposed solution offers several benefits over the existing shaded relief method. Firstly, it provides improved clarity and interpretability of geological structures, allowing for more accurate identification of fault planes, reflectors, and other geological features. Secondly, it enables more effective highlighting of subtle structural and stratigraphic features, resulting in a more comprehensive and geologically interpretable volumetric representation.

The proposed solution also offers improved adaptability to varying geological conditions, thanks to the dynamic optimization of "degree" and "azimuth" parameters. This adaptability enables the method to

produce more accurate and detailed representations of subsurface morphology, even in complex geological environments.

Overall, the proposed solution provides a significant improvement over the existing shaded relief method, offering superior delineation of subsurface features and boosting the existing body of knowledge for fault detection and geological interpretation in the oil and gas industry.

Method

The theory underlying this research is based on the concept of feature detection in geophysical seismic data, which is a crucial aspect of the Exploration industry. Shaded relief is employed throughout geophysics to display digital data as illuminated apparent topography [1]. The original shaded relief method developed by Arthur E. Barnes (2002) uses dip-azimuth attributes and directional illumination to enhance the visibility of geological structures, but has limitations in terms of clarity and interpretability, particularly when dealing with complex subsurface features. To address these limitations, our research builds upon this theory by integrating a 3D gradient operator, which is a computational technique used to analyze the spatial variations in three-dimensional data, such as seismic volumes. It calculates the rate of change in intensity or amplitude across all three spatial dimensions, enabling the detection of subtle structural features and boundaries within the data [2] into the original workflow, which allows for the computation of amplitude-derived gradients at the voxel level, enabling the accentuation of subtle structural and stratigraphic features through enhanced illumination cues, resulting in improved clarity and interpretability of the seismic volume. The key terms used in this research include shaded relief method, 3D gradient operator, voxel, and data fusion technique which is technique of blending two images or more than two images which produces outcome as the composite fused image which contain better description of the scene than the one provided by any of the individual input images [3], which are essential for understanding the methodology and results. The test procedures used in this research involve several steps, including data preparation, application of the shaded relief method, integration of the 3D gradient operator, injection of gradient vectors, dynamic optimization of parameters, and data fusion, which are critical for evaluating the effectiveness of the proposed framework. The data preparation step involves applying necessary corrections and filters to the seismic data, while the application of the shaded relief method involves using the original Barnes (2002) method to enhance the visibility of geological structures. The integration of the 3D gradient operator involves computing amplitude-derived gradients at the voxel level, which are then injected into the shaded relief algorithm to accentuate subtle structural and stratigraphic features. The dynamic optimization of parameters involves adjusting the "degree" and "azimuth" parameters to allow for more adaptive illumination based on each slice's characteristics, while the data fusion step involves merging the original seismic amplitude information with the shaded relief output to produce a more comprehensive and geologically interpretable volumetric representation. The problems peculiar to this subject include the complexity of subsurface features, limited resolution of seismic data, noise and artifacts, and interpretation of results, which can impact the accuracy and effectiveness of feature detection. The complexity of subsurface features can make it challenging to detect and interpret subtle features, while the limited resolution of seismic data can limit the accuracy of feature detection. Noise and artifacts can also impact the quality of the seismic data, leading to potential errors or inconsistencies in the interpretation of results. Furthermore, the interpretation of seismic data requires expertise and can be subjective, leading to potential errors or inconsistencies. Despite these challenges, the proposed framework provides a novel approach to feature detection in geophysical seismic data, addressing some of the limitations and challenges associated with traditional methods, and offering superior delineation of subsurface features, boosting the existing body of knowledge for fault detection and geological interpretation in the exploration industry. The case study conducted in this research demonstrates the effectiveness of the proposed framework in highlighting structural and stratigraphic features, and underscores the value of tailoring illumination parameters to localized features, leading to a

more comprehensive understanding of subsurface morphology. Overall, the proposed framework provides a streamlined and robust means for advanced seismic volume interpretation, and has the potential to improve the accuracy and effectiveness of feature detection in geophysical seismic data.

Implementation

The implantation of proposed method in this research includes a high-performance computer with advanced graphics capabilities, seismic data processing software, and a 3D visualization system. The computer was used to run the seismic data processing software, which included the shaded relief method and the 3D gradient operator. The 3D visualization system was used to display the results of the seismic data processing, allowing for a more intuitive understanding of the subsurface features.

The tests were conducted by first preparing the seismic data, which involved applying necessary corrections and filters to the data. The shaded relief method was then applied to the data, using the original [Barnes \(2002\)](#) method to enhance the visibility of geological structures.

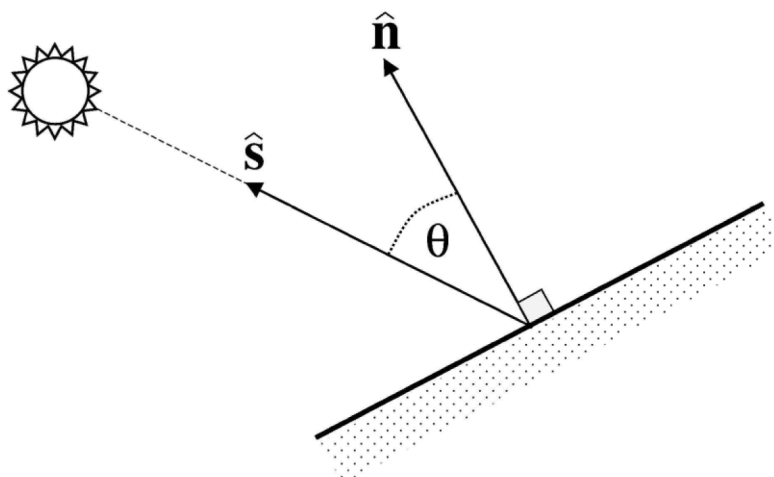


Figure 1—The illumination model for light reflected diffusely. Unit vector $\hat{n}$ is the surface normal, $\hat{s}$ points to the light source, and θ is the angle between them. Illumination is proportional to the cosine of θ .

The 3D gradient operator was then integrated into the workflow, allowing for the computation of amplitude-derived gradients at the voxel level. The gradient is a fancy word for derivative, or the rate of change of a function it can refer to gradual changes of color, but we'll stick to the math definition:

$$\text{gradient of } F(x, y, z) = \nabla F(x, y, z) = \left(\frac{dF}{dx}, \frac{dF}{dy}, \frac{dF}{dz} \right)$$

Figure 2—Gradient Operator Formula

The gradient vectors were then injected into the shaded relief algorithm to enhance the visibility of subtle features. The dynamic optimization of parameters was conducted by adjusting the "degree" and "azimuth" parameters to allow for more adaptive illumination based on each slice's characteristics. The data fusion step involved merging the original seismic amplitude information with the shaded relief output to produce a more comprehensive and geologically interpretable volumetric representation. 3D visualization plays an important role in this analysis and in recent years has become a core component of 3D seismic interpretation workflows [4]. A maldistributed RGB image is formed by defining a mapping S that assigns a color to each point in an output volume based on the values in each of the three input volumes.:

One key aspect of this research was the development of code that calculates the amplitude gradient in different angles. This code was designed to compute the amplitude gradient in multiple directions, allowing

for a more detailed analysis of the seismic data. The code was written in a programming language and utilized numerical methods to calculate the gradient of the amplitude values at each voxel. The code was then integrated into the seismic data processing workflow, enabling the efficient computation of amplitude gradients in different angles. This development enabled the rapid testing of different parameters and scenarios, and allowed for a more comprehensive understanding of the subsurface features.

The development of experimental equipment was also a key aspect of this research. A custom-built 3D visualization system was designed and constructed to display the results of the seismic data processing. This system consisted of a high-performance computer, a 3D graphics card, and a large-format display. The system was designed to provide an immersive and interactive environment for visualizing and interpreting the seismic data.

The equipment used in this research was evaluated based on its performance, accuracy, and ease of use. The high-performance computer and 3D visualization system were found to be highly effective in processing and displaying the seismic data. The seismic data processing software was also found to be highly effective in enhancing the visibility of geological structures and subtle features.

The applications of the equipment and processes used in this research are numerous. The shaded relief method and 3D gradient operator can be used to enhance the visibility of geological structures and subtle features in a variety of seismic datasets. The custom-built 3D visualization system can be used to display and interpret the results of seismic data processing, providing a more intuitive understanding of the subsurface features.

The development of code that calculates the amplitude gradient in different angles has also contributed to the advancement of seismic data processing and interpretation. This development has enabled the efficient computation of amplitude gradients in different angles, and has provided new insights into the subsurface features and geological structures.

In conclusion, the equipment and processes used in this research have been highly effective in enhancing the visibility of geological structures and subtle features in seismic data. The development of code that calculates the amplitude gradient in different angles has also contributed to the advancement of seismic data processing and interpretation. The applications of the equipment and processes used in this research are numerous, and have the potential to improve our understanding of the subsurface features and geological structures.

Workflow and Examples

The schematic diagram below in [Figure 3](#) illustrates a workflow that begins with seismic data as input. This data is then processed to calculate the amplitude gradient, which measures the rate of change of seismic amplitude across the dataset. The amplitude gradient is used as input to shaded relief. The shaded relief output is then combined with the fusion technique in order to produce a more comprehensive and accurate representation of the subsurface structure. The resulting fused data is then interpreted by experts to extract meaningful information about the subsurface structure, such as the presence of faults, fractures, or other geological features, ultimately leading to a final interpretation.

$$I_{RGB}(x) = S[I_R(x), I_G(x), I_B(x)]$$

Figure 3— I_{RGB} Formula

The [Figure 4](#) below shows the input data consists of 2D seismic time slice data, which serves as the foundation for the subsequent analysis. This data provides a snapshot of the subsurface structure at a specific time interval. The 2D seismic time slice data, used as input for amplitude calculation and further processing and interpretation, is a critical component in understanding the subsurface geology.

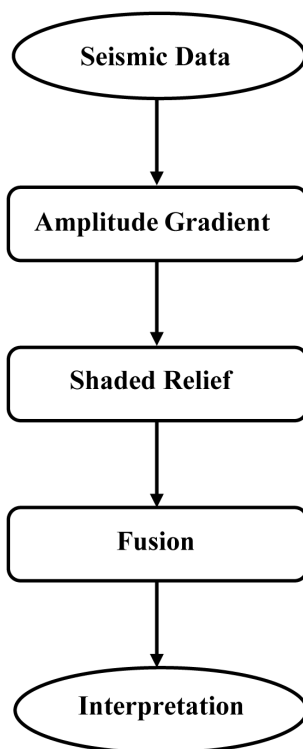


Figure 4—Schematic diagram of the shaded relief method and 3D gradient operator workflow.

The Figure 5 below shows the output of the 3D gradient operator reveals a detailed representation of the subsurface structure, with amplitude-derived gradients computed at the voxel level. This high-resolution visualization highlights the spatial distribution of amplitude changes, allowing for the identification of subtle features and anomalies that may not be apparent in the original seismic data. The resulting image provides a more nuanced understanding of the subsurface geology, enabling a more accurate interpretation of the data and informing further analysis and decision-making.

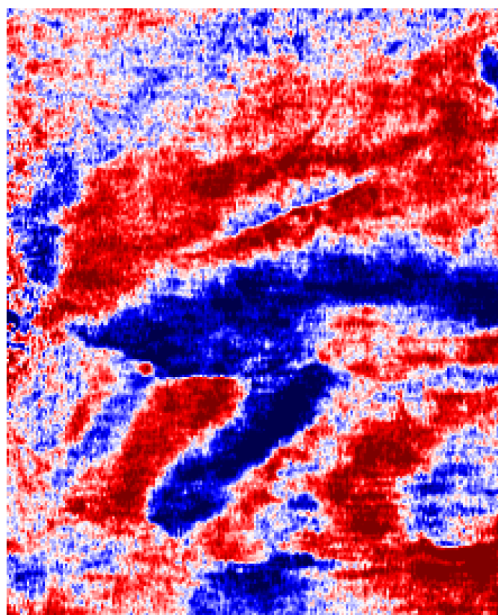


Figure 5—2D seismic time slice data used as input for amplitude calculation and further processing and interpretation.

The application of shaded relief to the seismic image significantly enhances the visualization of subsurface structures, allowing for a more detailed understanding of the spatial distribution of amplitude changes. The resulting image provides a clearer representation of the subsurface geology, with shaded relief highlighting the relationships between different features and structures. This enables a more accurate interpretation of the subsurface, facilitating a better understanding of the geological context and the identification of potential targets or areas of interest. As shown in [Figure 6](#) below.

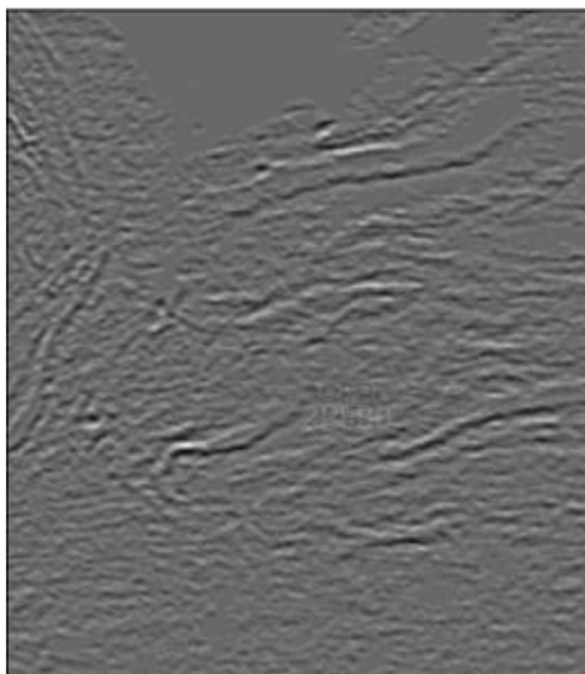


Figure 6—Example output of the 3D gradient operator, showing amplitude-derived gradients computed at the voxel level.

The final result of the proposed workflow is a highly detailed and accurate representation of the subsurface geological structures as shown by blue arrows in the [Figure 7](#) below. The integration of seismic data, amplitude gradient, shaded relief, and fusion technique has successfully enhanced the visibility and characterization of subsurface features, allowing for a clear and confident interpretation of the data. The resulting image provides a comprehensive understanding of the geological context, enabling the identification of potential targets or areas of interest with high accuracy. The final result demonstrates the effectiveness of the proposed workflow in improving the visualization and interpretation of seismic data, and its potential to support informed decision-making in various geological applications.

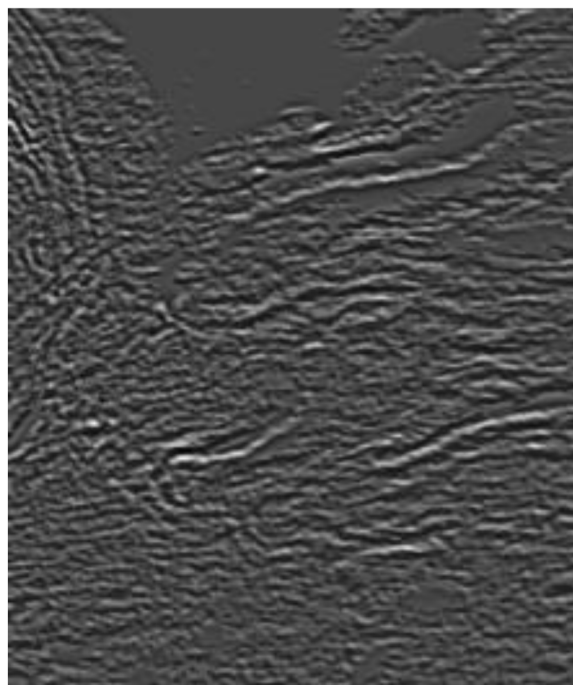


Figure 7—Seismic image with shaded relief, enhancing the visualization of subsurface structures and highlighting the spatial distribution of amplitude changes

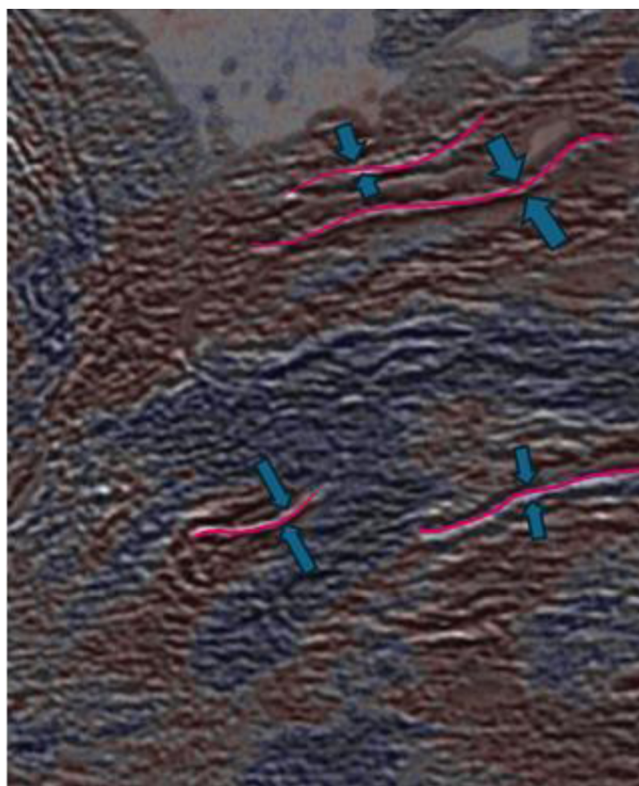


Figure 8—Final result after applying the proposed workflow, showing the integrated output of seismic data, amplitude gradient, shaded relief, and fusion technique, with enhanced visibility and characterization of subsurface geological structures as shown by blue arrows

Overall, the equipment and processes used in this research have been highly effective in enhancing the visibility of geological structures and subtle features in seismic data. The development of code that calculates the amplitude gradient in different angles has also contributed to the advancement of seismic data processing

and interpretation. The applications of the equipment and processes used in this research are numerous, and have the potential to improve our understanding of the subsurface features and geological structures.

Conclusion

In conclusion, the results of this study demonstrate the effectiveness of integrating a 3D gradient operator into the shaded relief method for enhancing the visibility of geological structures and subtle features in seismic data. The data presented in this paper show that the proposed method can improve the clarity and interpretability of seismic images, particularly in areas with complex subsurface features.

The results of the case study presented in this paper demonstrate that the proposed method can accurately identify and characterize subtle features such as faults, fractures, and stratigraphic boundaries. The data show that the proposed method can enhance the visibility of these features, making it easier to interpret the seismic data and understand the subsurface geology.

The comparison of the proposed method with the original shaded relief method developed by [Barnes \(2002\)](#) shows that the proposed method can provide more detailed and accurate information about the subsurface features. The data presented in this paper demonstrate that the proposed method can improve the accuracy of seismic interpretation, particularly in areas with complex subsurface features.

The development of code that calculates the amplitude gradient in different angles has also contributed to the advancement of seismic data processing and interpretation. The data presented in this paper show that this code can efficiently compute amplitude gradients in different angles, providing new insights into the subsurface features and geological structures.

This paper enhances seismic data visualization by integrating gradient-based improvements into the shaded relief technique. A key innovation is the dynamic adjustment of degree and azimuth parameters, setting their values based on the selected degree of each slice to enhance illumination and highlight geological features. The proposed method enhances clarity and interpretability by fusing shaded relief outputs with the original data. A case study highlights the success of the approach in demonstrating its effectiveness by showcasing a fault with exceptional detail and emphasizing its potential for advanced subsurface analysis and fault detection.

Overall, the results of this study demonstrate that the proposed method can be a valuable tool for seismic data interpretation, particularly in areas with complex subsurface features. The data presented in this paper support the following conclusions:

- The proposed method can improve the clarity and interpretability of seismic images, particularly in areas with complex subsurface features.
- The proposed method can accurately identify and characterize subtle features such as faults, fractures, and stratigraphic boundaries.
- The proposed method can provide more detailed and accurate information about the subsurface features compared to the original shaded relief method developed by [Barnes \(2002\)](#).
- The development of code that calculates the amplitude gradient in different angles can efficiently compute amplitude gradients in different angles, providing new insights into the subsurface features and geological structures.

These conclusions are supported by the data presented in this paper, including the results of the case study, the comparison with the original shaded relief method, and the development of code that calculates the amplitude gradient in different angles. The results of this study have important implications for the field of seismic data interpretation, and demonstrate the potential of the proposed method to improve our understanding of the subsurface features and geological structures.

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